Journal of Computational Physics 230 (2011) 4616–4635

Contents lists available at ScienceDirect

Journal of Computational Physics

journal homepage: www.elsevier.com/locate/jcp

A class of finite difference schemes with low dispersion and controllable dissipation for DNS of compressible turbulence

Zhen-Sheng Sun [a,b], Yu-Xin Ren [a,*], Cédric Larricq [a], Shi-ying Zhang [b], Yue-cheng Yang [b]

[a] Department of Engineering Mechanics, Tsinghua University, Beijing 100084, China
[b] Xi'an Research Institute of High-tech, Xi'an, Shaanxi 710025, China

ARTICLE INFO

Article history:
Received 10 June 2010
Received in revised form 22 February 2011
Accepted 24 February 2011
Available online 1 March 2011

Keywords:
Low dispersion scheme
Dissipation controllable scheme
WENO scheme
Hybrid scheme
Direct numerical simulation
Compressible turbulent flow

ABSTRACT

In this paper, a class of finite difference schemes which achieves low dispersion and controllable dissipation in smooth region and robust shock-capturing capabilities in the vicinity of discontinuities is presented. Firstly, a sufficient condition for semi-discrete finite difference schemes to have independent dispersion and dissipation is derived. This condition enables a novel approach to separately optimize the dissipation and dispersion properties of finite difference schemes and a class of schemes with minimized dispersion and controllable dissipation is thus obtained. Secondly, for the purpose of shock-capturing, one of these schemes is used as the linear part of the WENO scheme with symmetrical stencils to constructed an improved WENO scheme. At last, the improved WENO scheme is blended with its linear counterpart to form a new hybrid scheme for practical applications. The proposed scheme is accurate, flexible and robust. The accuracy and resolution of the proposed scheme are tested by the solutions of several benchmark test cases. The performance of this scheme is further demonstrated by its application in the direct numerical simulation of compressible turbulent channel flow between isothermal walls.

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1. Introduction

Turbulent flows are characterized by a large range of length scales. To be reliable, a direct numerical simulation (DNS) of such flows must resolve these scales, especially the small ones with accuracy in both amplitude and phase. Therefore, dissipation and dispersion properties of numerical schemes are of crucial importance. Because of their superior spectral properties, spectral methods and compact schemes are extensively used [1–5]. However, these methods are limited to compute flows without shock waves since they have been found to cause non-physical oscillations when applied to flow with discontinuities. ENO and WENO schemes [6,7], on the other hand, provide robust shock-capturing capability and high order accuracy. However, in their original forms, the dispersion and dissipation properties of these schemes are not optimized. The 5th WENO scheme of Jiang and Shu, for instance, is reported too dissipative for the detailed simulation of turbulent flow [8]. A lot of efforts have been therefore devoted to developing numerical schemes with high resolution and good shock-capturing capabilities.

The observations presented above suggest that, for designing such schemes, a natural choice is to combine the ENO/WENO scheme with another scheme with spectral-like resolution to form a so-called hybrid scheme. Adams and Shariff [9] proposed the hybrid compact-ENO scheme that coupled a non-conservative compact scheme with a shock-capturing ENO scheme for shock and turbulence interaction simulation. The compact scheme is applied where the flow field is smooth

* Corresponding author. Tel.: +86 10 62785543; fax: +86 10 62781824.
E-mail address: ryx@tsinghua.edu.cn (Y.-X. Ren).

0021-9991/$ - see front matter © 2011 Elsevier Inc. All rights reserved.
doi:10.1016/j.jcp.2011.02.038

and the ENO scheme is used near the discontinuities. Pirozzoli [10] derived a hybrid compact-WENO scheme in which a conservative compact scheme is coupled with a WENO scheme to make the overall scheme conservative. Ren et al. [11] improved the previous hybrid compact-WENO scheme by designing a continuous weight function to avoid the abrupt transition between the compact and WENO schemes. When solving the system of hyperbolic conservation laws, they used the characteristic decomposition to improve the resolution of the scheme. Shen and Yang [12] have also developed a hybrid compact-WENO scheme. Similar ideas have been adopted by Kim and Kwon [13], who proposed a high-order hybrid scheme which combined a central scheme and the WENO scheme for compressible flow field analysis. Costa and Don [14] developed the hybrid spectral-WENO scheme.

An alternative approach is to improve the spectral properties of the shock-capturing schemes by using the optimization techniques. The optimization procedures are usually based on the pioneering works of Tam and Webb [15], who devised the dispersion-relation-preserving (DRP) scheme for computational acoustics. Lockard et al. [16] developed an optimized ENO scheme for the solution of Euler equations. Their simulations showed that the optimized ENO scheme performed better than the non-optimized one for the case of linear wave propagation problems. Weirs and Candler [17] developed optimized WENO schemes to solve the hyperbolic conservation laws. They devised a strategy that consists of optimizing the weights of all candidate reconstructions and adding an additional candidate stencil to make the stencils symmetric rather than upwind-biased. This idea was further explored by Martín et al. [8], who developed a bandwidth-optimized WENO scheme for the DNS of compressible turbulent boundary layer. The essential idea of bandwidth-optimized WENO scheme is to determine the optimal weight by minimizing an elaborately designed integrated error function. Wang and Chen [18] proposed the optimized WENO scheme for solution of the linearized Euler equations with discontinuities. They designed a two level optimization procedure but only considered the biased stencils. Ponziani et al. [19] used the optimized WENO scheme for the DNS of isotropic compressible turbulence as well as aeroacoustic phenomena.

Concerning the optimization of the spectral properties of the finite differences schemes, it is generally accepted that the dispersion error should be minimized according to some chosen criteria. However, there are no general guidelines on how the dissipation should be optimized. A scheme with very large numerical dissipation is surely not suitable for DNS. On the other hand, Lechner et al. [20] noticed that the minimal dissipation produced by central difference scheme was insufficient in suppressing the numerical oscillation and could lead to instability. Therefore, a small amount of dissipation is needed to suppress numerical instabilities that may be caused by unresolved high wavenumber structures of the solution [20]. This point was further confirmed by Pirozzoli [10] who pointed out that a certain amount of dissipation was not necessarily a bad feature since, in the range of high wavenumbers, the waves propagated at an incorrect speed. It was then desirable to damp them as much as possible. The main difficulty in the optimization of the dissipation properties of the finite difference schemes is that the optimal dissipation is often problem dependent. For example, the bandwidth-optimized WENO scheme derived by Martín et al. [8] gave satisfying results in the DNS of supersonic boundary layer. However, this scheme is susceptible to cause numerical oscillations in some test cases as reported by Cai and Ladeinde [21].

According to the above discussion, it would be beneficial to design high order numerical schemes with minimized dispersion and controllable dissipation. However, this is not always possible since the dissipation and dispersion properties of a finite difference scheme are often inter-dependent. Indeed, in most of the available procedures for the optimization of the spectral properties of the finite difference schemes, the cost functions are the blending of the dissipation and dispersion errors. As a result, the change of the dissipation properties may deteriorate the already optimized dispersion properties. Therefore, it is desirable to have a class of finite difference schemes in which the dispersion and dissipation can be controlled separately.

In the present paper, we will design a class of semi-discrete finite difference schemes with minimized dispersion and controllable dissipation which is called the MDCD schemes hereafter. To design a MDCD scheme, a sufficient condition for the semi-discrete schemes to have independent dispersion and dissipation is derived, which makes it possible to optimize the dispersion properties of the scheme and to adjust the dissipation by the introduction of a free parameter. An important feature of the MDCD schemes is that the adjustment of the dissipation will not affect the optimized dispersion properties of the schemes. It is also found that the linear MDCD schemes can be used as the linear part of the WENO schemes to construct the so-called MDCD–WENO schemes which can be used to compute flow with discontinuities. As noted by Pirozzoli [22] and Shen et al. [23], the nonlinear mechanisms of the shock-capturing schemes may cause a dramatic corruption of their spectral properties even for the smooth flows. In the mean time, it is noted that the nonlinear weighting procedure of the WENO scheme is need only in the vicinity of discontinuities. Therefore, a hybrid scheme is proposed which is the blending of the linear MDCD scheme and the MDCD–WENO scheme using the technique of Ren et al. [11]. The approximate dispersion relation (ADR) [22] is computed to study the spectral properties of the nonlinear hybrid scheme and significant improvements over the original and optimized WENO schemes are observed. It is worthwhile to point out that the MDCD properties not only can improve the resolution of the finite difference schemes, but also enhance their flexibility and robustness. Because the dissipation of the MDCD schemes is adjustable, it is possible to make the MDCD schemes to posses smallest possible dissipation that is sufficient to ensure the stability and the damping of spurious numerical oscillations by adjusting only one parameter. On the other hand, for some very demanding problems with strong discontinuities, a rather large dissipation can be provided to enhance the robustness of the MDCD scheme so that a stable and converged numerical result can be obtained. Several benchmark test cases are presented in the present paper to demonstrate the superior performance of the proposed hybrid scheme. This scheme has also been applied successfully to the DNS of compressible turbulent channel flows.

2. Finite difference schemes with minimized dissipation and controllable dispersion

2.1. A sufficient condition for designing the MDCD schemes

In this section, a sufficient condition for a class of finite difference schemes to have independent dissipation and dispersion will be derived. We consider the scalar, one-dimensional hyperbolic conservation law given by

$$\frac{\partial u}{\partial t}+\frac{\partial f}{\partial x}=0, \tag{1}$$

where u is a conserved quantity and f is the flux. For simplicity, the semi-discrete finite difference scheme of Eq. (1)

$$\frac{\partial u}{\partial t}+f'_{\Delta}(x)=0 \tag{2}$$

is considered on uniform grids. The finite difference approximation to the first derivative $\frac{\partial f}{\partial x}$ based on $(2r+1)$ symmetrical grid points can be expressed as

$$f'_{\Delta}(x)=\frac{1}{\Delta x}\sum_{m=-r}^{r} b_m f_m, \tag{3}$$

where $f_m = f(x+m\Delta x)$. It is well-known that Eq. (3) can achieve $2r$th order of accuracy when the coefficients b_m are properly chosen. However, the main drawback of such an approach is that the dispersion and dissipation properties of the resulting scheme are not optimal. On the other hand, spectral-optimized finite difference schemes such as the DRP schemes optimize the dissipation and dispersion properties at the expense of formal order of accuracy. This is also the approach of the present paper. Furthermore, as discussed in Section 1, we want to design a class of finite difference schemes with minimized dispersion and controllable dissipation. For this purpose, the following lemma is introduced.

Lemma 1. *If $f'_{\Delta}(x)$ in Eq. (3) approximates $\frac{\partial f}{\partial x}$ to $(2r-2)$th order of accuracy, the dispersion and dissipation of the corresponding semi-discrete scheme, Eq. (2), are determined respectively by two free parameters γ_{disp} and γ_{diss} which are independent of each other.*

Proof. If $f'_{\Delta}(x)$ approximates $\frac{\partial f}{\partial x}$ to $(2r-2)$th order of accuracy, it can be written in the following general form,

$$f'_{\Delta}(x)=f'_{\Delta C}+\alpha f'_{\Delta L}+\beta f'_{\Delta R}, \tag{4}$$

where

$$f'_{\Delta C}=\frac{1}{\Delta x}\sum_{m=-r+1}^{r-1} a_m f_m=\frac{\partial f}{\partial x}+O(\Delta x)^{2r-2},$$

$$f'_{\Delta L}=\frac{1}{\Delta x}\sum_{m=-r}^{r-1}(-1)^{r-m-1}\binom{2r-1}{r-m-1}f_m,$$

$$f'_{\Delta R}=\frac{1}{\Delta x}\sum_{m=-r+1}^{r}(-1)^{r-m}\binom{2r-1}{r-m}f_m,$$

and the binomial coefficient $\binom{n}{k}$ is defined by

$$\binom{n}{k}=\frac{n!}{k!(n-k)!}.$$

Since $f'_{\Delta C}$ is a $(2r-2)$th order approximation to $\frac{\partial f}{\partial x}$using $(2r-1)$ symmetrical points, its coefficients, a_m, can be uniquely determined and the solutions are

$$a_m=\begin{cases}\frac{\prod_{q=-r+1,q\neq m,0}^{r-1}(-q)}{\prod_{k=-r+1,k\neq m}^{r-1}(m-k)}, & \text{if } m\neq 0,\\ 0, & \text{otherwise.}\end{cases}$$

Furthermore, it is easy to prove that

$$f'_{\Delta L}=\frac{1}{\Delta x}\Delta^{2r-1}f_{-r}\sim O(\Delta x^{2r-2})$$

and

$$f'_{\Delta R}=\frac{1}{\Delta x}\nabla^{2r-1}f_{r}\sim O(\Delta x^{2r-2}),$$

where $\Delta f_{-r} = f_{-(r-1)} - f_{-r}$ and $\nabla f_r = f_r - f_{r-1}$. Therefore, Eq. (4) is a $(2r-2)$th order approximation of $\frac{\partial f}{\partial x}$ with two free parameters α and β. The dispersion and dissipation properties of the semi-discrete scheme corresponding to Eq. (4) can be deduced as follows. Considering $f(x)$ to be a pure harmonic function

$$f(x) = e^{i\omega x}, \tag{5}$$

we have

$$\frac{\partial f}{\partial x} = ik/\Delta x e^{i\omega x}, \tag{6}$$

where $k = \omega\Delta x$ is the scaled wavenumber with $0 \leqslant k \leqslant \pi$. Substituting Eq. (5) into Eq. (4), we obtain

$$f'_\Delta(x) = ik'/\Delta x e^{i\omega x}. \tag{7}$$

In Eq. (7), k' is the modified wavenumber given by

$$k' = \Re(k') + i\Im(k'), \tag{8}$$

where $\Re(k')$ and $\Im(k')$ represent the real and imaginary part of k' which are associated with the dispersion and dissipation of the scheme, respectively. Their specific forms are

$$\Re(k') = \gamma_{disp}\sin rk + \sum_{m=1}^{r-1}\left[2a_m + (-1)^{r-m-1}\gamma_{disp}\left(\binom{2r-1}{r-m-1} - \binom{2r-1}{r-m}\right)\right]\sin mk, \tag{9}$$

$$\Im(k') = \gamma_{diss}\left[\cos rk + (-1)^r\binom{2r-1}{r} + \sum_{m=1}^{r-1}(-1)^{r-m}\cos mk\binom{2r}{r-m}\right], \tag{10}$$

where

$$\gamma_{disp} = \alpha + \beta, \quad \gamma_{diss} = \alpha - \beta. \tag{11}$$

It is apparent that γ_{disp} and γ_{diss} are two independent parameters. This completes the proof. □

Lemma 1 can be generalized if it is allowed to further reduce the order of accuracy of the scheme. In fact, the procedure for proving Lemma 1 can be also applied to $f'_{\Delta C}$ in Eq. (4) with a reduced order of accuracy. Repeating this procedure, the following lemma can be obtained.

Lemma 2. *If f'_Δ in Eq. (3) approximates $\frac{\partial f}{\partial x}$ to $(2r-2n)$th order of accuracy $(1 \leqslant n < r)$, the dispersion and dissipation of the corresponding semi-discrete scheme are determined separately by two sets of free parameters each with n elements.*

Lemmas 1 and 2 make it possible to design the finite difference schemes with the MDCD properties. To be specific, we only consider Lemma 1 in the present paper. In this case, γ_{disp} can be determined by optimization of the dispersion properties. γ_{diss} is chosen to provide the dissipation needed in a specific simulation. After the determination of γ_{diss} and γ_{disp}, α and β are computed according to Eq. (11) and the corresponding MDCD scheme is thus constructed.

2.2. Spectral properties of the MDCD scheme

To facilitate the construction of the MDCD scheme, the spectral properties and the optimization procedures of the finite difference scheme will be studied in detail for the case $r = 3$ in this subsection. When $r = 3$, the specific form of Eq. (4) is

$$f'_\Delta(x) = \frac{1}{\Delta x}\begin{bmatrix} \left(-\frac{1}{2}\gamma_{disp} - \frac{1}{2}\gamma_{diss}\right)f_{-3} + \left(2\gamma_{disp} + 3\gamma_{diss} + \frac{1}{12}\right)f_{-2} \\ +\left(-\frac{5}{2}\gamma_{disp} - \frac{15}{2}\gamma_{diss} - \frac{2}{3}\right)f_{-1} + 10\gamma_{diss}f_0 + \left(\frac{5}{2}\gamma_{disp} - \frac{15}{2}\gamma_{diss} + \frac{2}{3}\right)f_1 \\ +\left(-2\gamma_{disp} + 3\gamma_{diss} - \frac{1}{12}\right)f_2 + \left(\frac{1}{2}\gamma_{disp} - \frac{1}{2}\gamma_{diss}\right)f_3 \end{bmatrix}, \tag{12}$$

which is a 4th order approximation of $\frac{\partial f}{\partial x}$. The real part of the modified wavenumber of Eq. (12) can be written as

$$\Re(k') = \left(\frac{4}{3} + 5\gamma_{disp}\right)\sin k + \left(-\frac{1}{6} - 4\gamma_{disp}\right)\sin 2k + \gamma_{disp}\sin 3k. \tag{13}$$

The optimized value of γ_{disp} can be evaluated by minimizing the following integrated error function [8]

$$E = \frac{1}{e^{\upsilon\pi}}\int_0^\pi e^{\upsilon(\pi-k)}(\Re(k') - k)^2 dk. \tag{14}$$

We note that υ in Eq. (14) is a parameter which is used to control the relative importance of the low wavenumber and high wavenumber errors. The relationship between E and γ_{disp} is plotted in Fig. 1 for different values of υ. It can be easily seen that

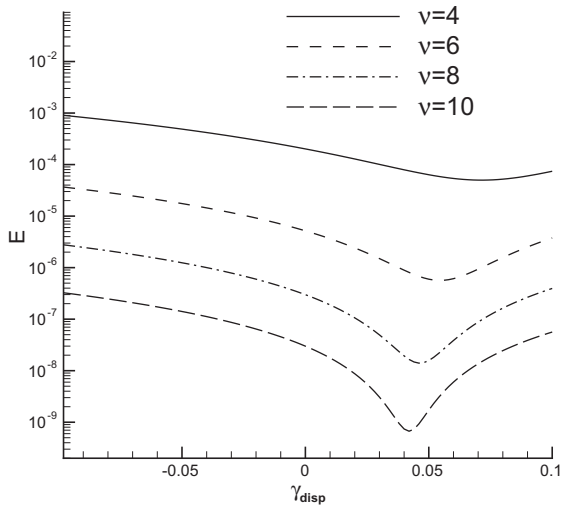


Fig. 1. Relation between E and γ_{disp} for different υ.

Table 1
Optimized values of γ_{disp} due to different υ.

υ	4	6	8	10
γ_{disp}	0.0714071	0.0545455	0.0463783	0.0420477

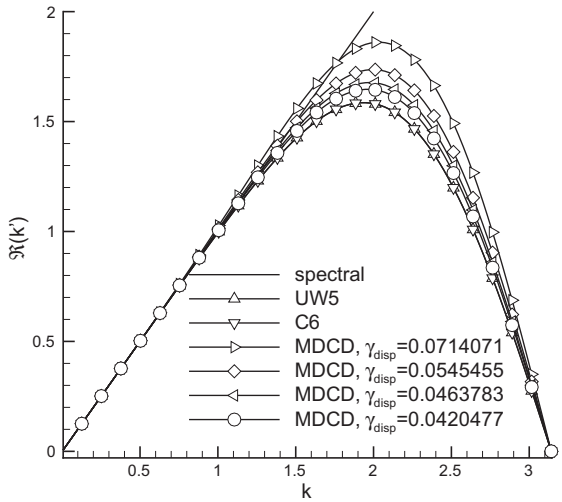


Fig. 2. Comparison of dispersion properties of various schemes. Note the dispersion properties of UW5 and C6 are the same.

E has a minimum and the corresponding value of γ_{disp} is the optimized one. The optimal values of γ_{disp} corresponding to different values of υ are shown in Table 1. The dispersion properties of the 4th order MDCD schemes based on the optimized γ_{disp} for different υ are shown in Fig. 2. For the purpose of comparison, the dispersion properties of the 5th order upwind-biased scheme (UW5, the original WENO scheme disabling the nonlinear weights) and the 6th order central difference scheme (C6) are also shown. It can be seen that the dispersion properties of the UW5 is the same as that of C6 while the MDCD schemes show higher resolutions. To show the dispersion properties more clearly, the dispersion errors defined by $\Re(k')/k - 1$ are plotted in Fig. 3. The final optimized value of γ_{disp} for the practical applications will be determined in the next section where the nonlinear mechanism needed for shock-capturing will also be considered.

The dissipation of the MDCD scheme can be adjusted by changing the value of γ_{diss}. To ensure the stability of the scheme, the dissipation should be non-negative for all wavenumbers. The imaginary part of the modified wavenumber of Eq. (12) can be written as:

$$\Im(k') = \gamma_{diss} g(cos(k)), \tag{15}$$

where

$$g(x) = 4x^3 - 12x^2 + 12x - 4, \quad -1 \leqslant x \leqslant 1.$$

As $g'(x) = 12(x-1)^2 \geqslant 0$, $g(x)$ is non-decreasing. It is straightforward to verify $g(1) = 0$. Therefore, we have $g(x) \leqslant 0$ for $-1 \leqslant x \leqslant 1$. This proves $g(cos(k)) \leqslant 0$. If $f'(u) \geqslant 0$, the dissipation of the MDCD scheme is non-negative as long as $\gamma_{diss} \geqslant 0$. For the case $f'(u) \leqslant 0$, $\gamma_{diss} \leqslant 0$ will ensure the dissipation is non-negative. It is clear that the dissipation of the MDCD scheme is an increasing function of $|\gamma_{diss}|$. From this point, we use the notation MDCD ($|\gamma_{diss}|$) to denote a MDCD scheme with a specific dissipation corresponding to the value of γ_{diss}. For example, the dissipation of MDCD(0) and MDCD(0.05) are shown in Fig. 4 where the dissipations of UW5 and C6 are also depicted. If we use

$$G = \left| \int_0^{\pi} \Im(k') dk \right| \tag{16}$$

as a measure of the dissipation, it is clear that G is a linear function of $|\gamma_{diss}|$. The relation between G and $|\gamma_{diss}|$ is shown in Fig. 5 where it can be seen that the dissipation error of the MDCD scheme can be chosen in a range from zero to an infinite

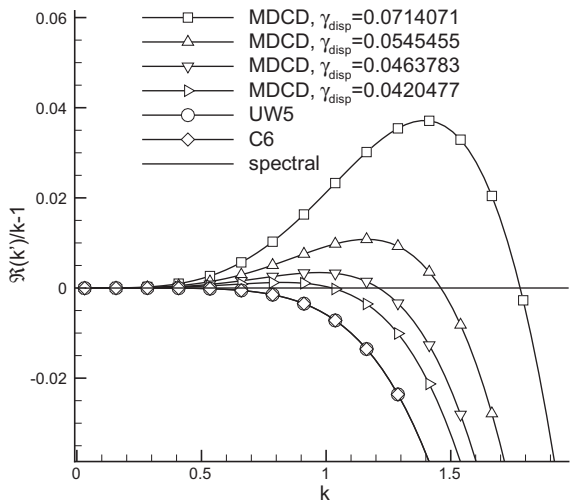


Fig. 3. Comparison of dispersion errors of various schemes. Note the dispersion errors of UW5 and C6 are the same.

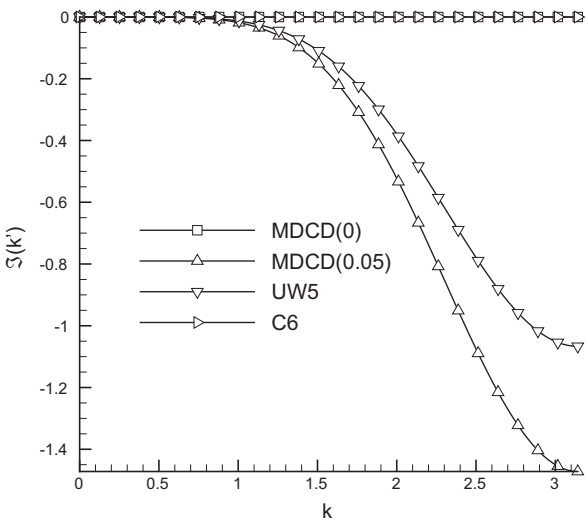


Fig. 4. Comparison of the dissipation of various schemes.

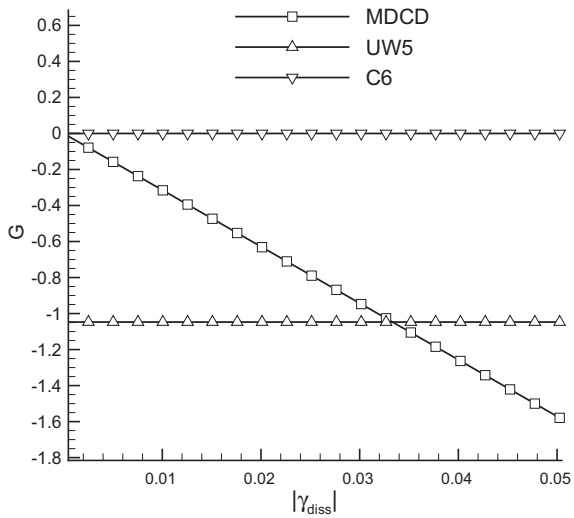


Fig. 5. Relation between G and $|\gamma_{diss}|$.

large value. On the other hand, the dissipation errors of UW5 and C6 are fixed. It can be also observed that the dissipation of MDCD(0.035) is similar to that of UW5 while both MDCD(0) and C6 have zero dissipation.

3. The nonlinear MDCD scheme

3.1. The MDCD–WENO scheme

The semi-discrete finite difference scheme of Eq. (1) in conservative form can be written as

$$\frac{du_j}{dt} = L(u)_j = -\frac{1}{\Delta x}(\hat{f}_{j+1/2} - \hat{f}_{j-1/2}), \tag{17}$$

where $\hat{f}_{j+1/2}$is the numerical flux. To achieve shock-capturing capability, we rewrite the numerical flux of the MDCD scheme within the framework of the WENO scheme with symmetrical stencils. That is

$$\hat{f}_{j+1/2} = \sum_{k=0}^{r} C_k^r q_k^r(x_{j+1/2}) + \sum_{k=0}^{r} (\omega_r - C_k^r) q_k^r(x_{j+1/2}), \tag{18}$$

where $q_k^r(x_{j+1/2})$ is the rth order accurate approximation of the numerical flux based on a set of candidate stencils denoted by $S_k = (x_{j+k-r+1}, \ldots, x_{j+k})$. The detailed formulation of $q_k^r(x_{j+1/2})$ can be found in [8]. Using the notation of [7], C_k^r is the optimal weight and ω_k is the weight related to the relative smoothness of f on each set of candidate stencils. In terms of the numerical flux in Eq. (18), the spatial derivative in Eq. (4) is evaluated as

$$f'_\Delta(x_j) = \frac{1}{\Delta x}\left[\sum_{k=0}^{r} C_k^r \left(q_k^r(x_{j+1/2}) - q_k^r(x_{j-1/2})\right) + \Omega_{j+1/2} - \Omega_{j-1/2}\right], \tag{19}$$

where

$$\Omega_{j+1/2} = \sum_{k=0}^{r} (\omega_r - C_k^r) q_k^r(x_{j+1/2}). \tag{20}$$

According to Eq. (19), the finite difference approximation of the derivative can be split into two parts

$$f'_\Delta(x_j) = f'_{\Delta-linear} + f'_{\Delta-nonlinear}. \tag{21}$$

The linear part

$$f'_\Delta\text{-linear} = \frac{1}{\Delta x}\left[\sum_{k=0}^{r} C_k^r \left(q_k^r(x_{j+1/2}) - q_k^r(x_{j-1/2})\right)\right] \tag{22}$$

mainly affects the formal order of accuracy of the scheme and the nonlinear part

$$f'_\Delta\text{-nonlinear} = \frac{1}{\Delta x}[\Omega_{j+1/2} - \Omega_{j-1/2}] \tag{23}$$

is responsible for capturing the discontinuities.

Because both Eqs. (4 and 22) are based on the same symmetric stencils, C_k^r can be written as functions of γ_{disp} and γ_{diss} (or α and β). When $r = 3$, the relations are shown in Table 2. It is noted that the corresponding MDCD–WENO scheme is 4th order accurate and satisfies the MDCD properties. We also note that both the linear and nonlinear parts of the scheme are important in order to design a finite difference with good spectral properties and shock-capturing capability. The nonlinear weight ω_r in the present paper is that proposed by Taylor et al. [25], which is proved to be superior to the original weight of Jiang and Shu [7].

3.2. The hybrid scheme

The nonlinear finite difference scheme MDCD–WENO has good shock-capturing capabilities. However, the spectral properties of the shock-capturing schemes may be much different from their linear counterpart. In other words, the nonlinear mechanisms of the shock-capturing schemes may deteriorate the spectral properties of the schemes even in the smooth region [22,23]. Moreover, the computation of the nonlinear terms of the MDCD–WENO scheme is very time-consuming. To overcome these limitations, a hybrid scheme combining MDCD–WENO with its linear counterpart is developed. The linear MDCD scheme is applied in the smooth region while the nonlinear mechanism is activated in the vicinity of discontinuities. In the present paper, the hybridization is based on the method of Ren et al. [11] to avoid the abrupt transition from the linear scheme to the nonlinear one. The numerical flux of the hybrid scheme is given by

$$\hat{f}_{j+1/2} = \sigma_{j+1/2}\hat{f}^{\mathrm{MDCD}}_{j+1/2} + (1-\sigma_{j+1/2})\hat{f}^{\mathrm{MDCD-WENO}}_{j+1/2}, \tag{24}$$

where $\sigma_{j+1/2}$ is the weight, $\hat{f}^{\mathrm{MDCD}}_{j+1/2}$ and $\hat{f}^{\mathrm{MDCD-WENO}}_{j+1/2}$ represent the numerical fluxes computed by the linear MDCD scheme and the MDCD–WENO scheme respectively. The weight is a continuous function of the smoothness indicator and is designed to be [11]

$$\sigma_{j+1/2} = \min\left(1, \frac{r_{j+1/2}}{r_c}\right). \tag{25}$$

In Eq. (25), $r_{j+1/2}$ is the smoothness indicator computed by

$$r_{j+1/2} = \min(r_j, r_{j+1}), \tag{26}$$

where

$$r_j = \frac{|2\Delta f_{j+1/2}\Delta f_{j-1/2}| + \varepsilon}{(\Delta f_{j+1/2})^2 + (\Delta f_{j-1/2})^2 + \varepsilon} \tag{27}$$

and $\Delta f_{j+1/2} = f_{j+1} - f_j$. The positive parameter ε is given by

$$\varepsilon = \frac{0.9r_c}{1-0.9r_c}\xi^2, \tag{28}$$

where ξ is chosen as $\xi = 10^{-3}$. r_c in Eq. (25) is a threshold for identifying the smooth flow. When $r_{j+1/2} \geqslant r_c$, the flow is considered to be smooth and the MDCD–WENO flux is not computed to improve the efficiency of the hybrid scheme. In the present paper, r_c is set to be $r_c = 0.4$. By adopting a fixed value of r_c, the dissipation of the numerical scheme is only controlled by γ_{diss}.

3.3. The spectral properties of the hybrid scheme

For a nonlinear scheme, no analytical formula of the spectral relations can be obtained. However, the modified wavenumber of the nonlinear scheme in terms of the approximate dispersion relation (ADR) can be obtained using the technique of Pirozzoli [22]. In order to optimize the dispersion properties of the proposed hybrid scheme (referred to as MDCD-HY hereafter), its ADR based on the optimized γ_{disp} presented in Table 1 is plotted in Fig. 6. The dispersion properties of the original WENO scheme developed by Jiang and Shu [7](referred to as WENO-JS hereafter) and the optimized WENO scheme developed by Martín et al. [8] (referred to as WENO-SYMBO hereafter) are also shown for comparison. The selection of the γ_{disp} for the MDCD-HY scheme is based on the following criterion: the resolving efficiency [24] $\xi = \frac{k_c}{\pi}$ should be maximized subjected to the constraint $|\Re(k')/k - 1| \leqslant \tau$ for all $k \leqslant k_c$, where τ is the tolerance of the dispersion error. In [17], $\tau = 1.5\%$ is used in the optimization procedure. Based on this tolerance, the optimal γ_{disp} is $\gamma_{disp} = 0.0545455$ ($\upsilon = 6$) according to Fig. 7. However, it

Table 2
Optimized weights $C_k (0 \leqslant k \leqslant 3)$ for the MDCD–WENO scheme. The superscript r is omitted for brevity.

C_0	C_1	C_2	C_3
$\frac{3}{2}(\gamma_{disp} + \gamma_{diss})$	$\frac{1}{2} - \frac{3}{2}\gamma_{disp} + \frac{9}{2}\gamma_{diss}$	$\frac{1}{2} - \frac{3}{2}\gamma_{disp} - \frac{9}{2}\gamma_{diss}$	$\frac{3}{2}(\gamma_{disp} - \gamma_{diss})$

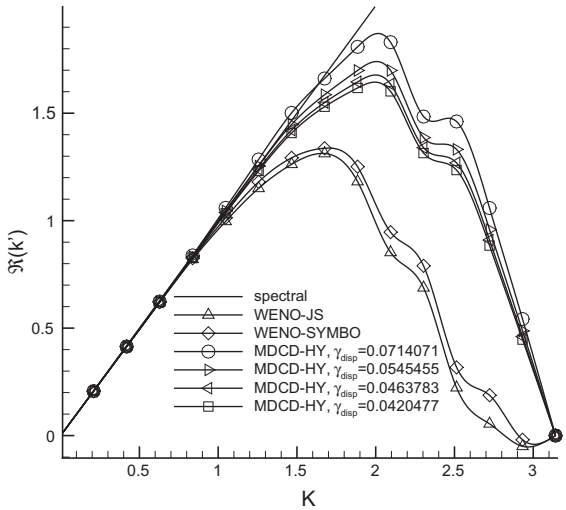


Fig. 6. Comparison of dispersion properties of various shock-capturing schemes.

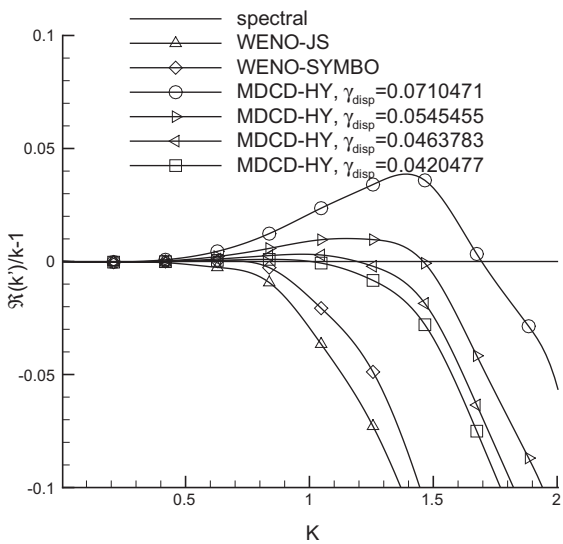


Fig. 7. Comparison of dispersion errors of various shock-capturing schemes.

can be seen in Fig. 7 that the corresponding dispersion error is relatively large in the region $0.5 \leqslant k \leqslant 1.5$. To reduce the dispersion error for the low wavenumber ($k \leqslant 1.5$) elements of the solution, we choose $\tau = 0.5\%$ in the present and the optimal value of γ_{disp} is

$$\gamma_{disp} = 0.0463783 \tag{29}$$

which is corresponding to $\upsilon = 8$. The resolving efficiency of the resulting MDCD-HY scheme is 41%. Based on the same tolerance, the resolving efficiency of WENO-JS and WENO-SYMBO are 21% and 28%, respectively. According to these results, a significant improvement in the resolution of MDCD-HY over WENO-JS and WENO-SYMBO can be expected.

The dissipation of the hybrid scheme is function of γ_{diss}, when $f'(u) \geqslant 0$, we need $\gamma_{diss} \geqslant 0$ to ensure that the dissipation is non-negative. Furthermore, $C_k (0 \leqslant k \leqslant 3)$ is usually required to be non-negative, which leads to $\gamma_{diss} \leqslant \gamma_{disp}$. Therefore, we have

$$0 \leqslant \gamma_{diss} \leqslant 0.0463783. \tag{30}$$

When $f'(u) \leqslant 0$, the corresponding constraint is

$$-0.0463783 \leqslant \gamma_{diss} \leqslant 0. \tag{31}$$

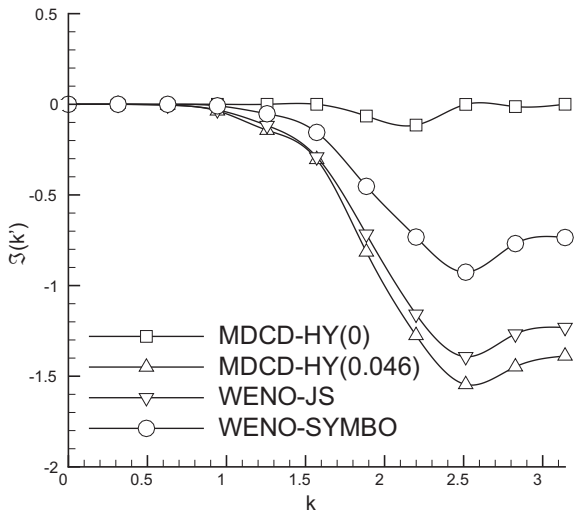


Fig. 8. Comparison of dissipation properties of various shock-capturing schemes.

We use the notation MDCD-HY ($|\gamma_{diss}|$) to denote a MDCD-HY scheme with the specific dissipation corresponding to the value of γ_{diss}. The dissipation properties of the MDCD-HY(0) and MDCD-HY(0.046) are computed using the technique of ADR [22] and are shown in Fig. 8. In principle, the dissipation must be as small as possible but large enough to ensure the stability of the simulation and the damping of spurious numerical oscillations. Consequently, the optimal value of dissipation is problem-dependent. A significant advantage of the MDCD-HY scheme is that it provides the flexibility in adjusting the dissipation without corrupting its dispersion properties. When numerical stability is not an issue and there are no spurious oscillations, the dissipation error of MDCD-HY produced by the optimal weight and nonlinear mechanism can be almost eliminated by choosing a γ_{disp} with a small absolute value. On the other hand, MDCD-HY can also supply large dissipation by increasing $|\gamma_{diss}|$. As shown in Fig. 8, the largest dissipation of MDCD-HY can be actually larger than that of WENO-JS. The flexibility in adjusting the dissipation makes it possible to develop a versatile and robust code for the simulation of complex problems with discontinuities. In principle, γ_{diss} can be also chosen in an adaptive way. For example, the γ_{diss} of the MDCD-HY scheme at a grid point can be related to the distribution of local numerical solutions in the vicinity of this grid point. This point will be further investigated in the future work.

3.4. Application to Euler and Navier–Stokes equations

In this section, the MDCD-HY scheme is applied to solve Euler and Navier–Stokes equations. For brevity, the procedure is illustrated using one-dimensional Euler equations of gas dynamics, which, in conservation form, can be written as

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} = \mathbf{0}, \tag{32}$$

where

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho u \\ \rho E \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho u H \end{bmatrix}.$$

The inviscid numerical fluxes are computed by considering the local characteristic flux decompositions. The procedure involves the following steps:

(1) At each fixed $x_{j+1/2}$, the average state $\mathbf{U}_{j+1/2}$ is determined by a simple average or the Roe average.
(2) The left eigenvector matrix $\mathbf{L}^{(i)}_{j+1/2}$ $(i = 1, 2, 3)$ and right eigenvector matrix $\mathbf{R}^{(i)}_{j+1/2}$ $(i = 1, 2, 3)$ as well as the eigenvalues $\lambda^{(i)}_{j+1/2}$ $(i = 1, 2, 3)$ of the Jacobian matrix $\partial\mathbf{F}/\partial\mathbf{U}$ are evaluated at the average state.
(3) For each set of candidate stencils, the local characteristic decomposition of fluxes are computed by

$$w^{(i)}_m = \mathbf{L}^{(i)}_{j+1/2}\mathbf{F}_m, \quad m \in [j - r + 1, j + r],$$

where $r = 3$ is chosen in the present paper.

(4) The numerical characteristic flux functions $\hat{w}_{j+1/2}^{(i),\pm,\mathrm{MDCD}}$ or $\hat{w}_{j+1/2}^{(i),\pm,\mathrm{MDCD-WENO}}$ are computed from the local characteristic fluxes $w_m^{(i),\pm}$ by using the linear MDCD scheme or the nonlinear MDCD–WENO scheme. $w_m^{(i),\pm}$ can be evaluated by the Roe-type local characteristic flux according to

$$\begin{cases} w_m^{(i),+,Roe} = w_m^{(i)} \\ w_m^{(i),-,Roe} = 0 \end{cases} \quad \text{if } \lambda_{j+1/2}^{(i)} > 0,$$

$$\begin{cases} w_m^{(i),+,Roe} = 0 \\ w_m^{(i),-,Roe} = w_m^{(i)} \end{cases} \quad \text{if } \lambda_{j+1/2}^{(i)} < 0$$

or the flux based on local Lax-Friedrichs flux splitting which can be written as

$$\begin{cases} w_m^{(i),+,LF} = \dfrac{\mathbf{L}_{j+1/2}^{(i)}\mathbf{F}_m + \max\limits_{l\in(j-r+1,\ldots,j+r)} |\lambda_l^{(i)}| \mathbf{L}_{j+1/2}^{(i)}\mathbf{U}_m}{2}, \\ w_m^{(i),-,LF} = \dfrac{\mathbf{L}_{j+1/2}^{(i)}\mathbf{F}_m - \max\limits_{l\in(j-r+1,\ldots,j+r)} |\lambda_l^{(i)}| \mathbf{L}_{j+1/2}^{(i)}\mathbf{U}_m}{2}. \end{cases}$$

We basically use the Roe-type scheme as it is less dissipative and thus achieves higher resolution. However, it should be noted that the Roe-type WENO schemes admit rarefaction shocks that do not satisfy the entropy condition and an entropy fix procedure with

$$w_{j+1/2}^{(i),\pm} = \begin{cases} w_{j+1/2}^{(i),\pm Roe} & \text{if } \lambda_j^{(i)}\lambda_{j+1}^{(i)} > 0, \\ w_{j+1/2}^{(i),\pm LF} & \text{otherwise} \end{cases}$$

is used in the practical computation.

(5) The approach of hybridization is the same as that previously used for the scalar conservation law. That is

$$\hat{w}_{j+1/2}^{(i),\pm} = \sigma_{j+1/2}^{(i)} w_{j+1/2}^{(i),\pm,\mathrm{MDCD}} + (1-\sigma_{j+1/2}^{(i)}) w_{j+1/2}^{(i),\pm,\mathrm{MDCD-WENO}} \tag{33}$$

with

$$\sigma_{j+1/2}^{(i)} = \min\left(1, \ \frac{r_{j+1/2}^{(i)}}{r_c}\right),$$

$$r_{j+1/2}^{(i)} = \min\left(r_j^{(i)}, \ r_{j+1}^{(i)}\right),$$

and

$$r_j^{(i)} = \frac{\left|2\Delta w_{j+1/2}^{(i)}\Delta w_{j-1/2}^{(i)}\right| + \varepsilon}{(\Delta w_{j+1/2}^{(i)})^2 + (\Delta w_{j-1/2}^{(i)})^2 + \varepsilon}.$$

(6) The inviscid numerical flux is finally computed by

$$\widehat{\mathbf{F}}_{j+1/2} = \mathbf{R}_{j+1/2}\left(\widehat{\mathbf{W}}_{j+1/2}^{+} + \widehat{\mathbf{W}}_{j+1/2}^{-}\right).$$

The MDCD-HY scheme for the one-dimensional Euler equation can be extended to solve multi-dimensional Euler and Navier–Stokes equations on Cartesian and curvilinear coordinates. The detailed formulation is omitted here and further details can be found in [26] where the hybrid compact-WENO scheme on curvilinear coordinates is considered.

4. Numerical tests

In order to analyze the behavior of the MDCD-HY scheme for solving problems characterized by a wide range of length scales, various test problems have been computed, including a wave packet with a broadband sine waves, shock/entropy wave interaction and an acoustic disturbance characterized by different length scales. For comparison, the computations have also been carried out with WENO-JS and WENO-SYMBO. For all one-dimensional problems presented in this section, the MDCD-HY(0) can achieve stable numerical solutions. Nevertheless, in some test cases, results obtained using MDCD-HY with larger dissipation are also presented to show the influence of the numerical dissipation. To demonstrate the robustness and computational efficiency of the proposed scheme, a two-dimensional viscous shock tube problem with different initial conditions has been solved. As a further test of the present scheme, the DNS of the compressible turbulent flow between isothermal walls is presented in this section. For all one and two-dimensional test cases, the low storage fourth order Runge–Kutta scheme [27] is used for the time integration. For the three-dimensional DNS, an implicit dual-time stepping LU-SGS algorithm [28] is adopted to advance the solution in time.

4.1. Transportation of a wave packet

The linear advection equation is used to test the accuracy of the MDCD-HY scheme. The governing equation is

$$\frac{\partial u}{\partial t}+\frac{\partial u}{\partial x}=0.$$

The initial condition is $u_0(x)=\frac{1}{m}\sum_{k=1}^{m}\sin(2\pi kx)$, and periodic boundary conditions are imposed at the two end points of the region $0\leqslant x\leqslant 1$. The solution of this problem is characterized by the wave packet with different length scales. As m becomes larger, it contains higher wavenumber elements. The simulation is carried out up to t = 1.0 using a Courant number CFL = 0.1. Four cases with m = 1, m = 5, m = 10 and m = 20 are computed on different grid points. Errors of the numerical solutions in L_2 norm are shown in Fig. 9. The results indicate that WENO-JS, WENO-SYMBO and MDCD-HY can all achieve their theoretical order of accuracy (5th, 3th and 4th order, respectively) when the grid number is large enough. Because WENO-JS has the highest order of accuracy, it will gradually become the most accurate scheme as the increase of the grid numbers. However, this behavior is in close relation with the length scale of the wave packet. When m = 1, 5, 10 and 20, at least 80, 500, 1000 and 1500 grid points are needed, respectively to make WENO-JS more accurate than MDCD-HY. This is corresponding to 75–100 grid points in one wavelength to resolve the highest wavenumber component of the solution. In practice applications such as DNS of turbulent flow, one can rarely use such a large number of grid points. If about 10–20 points in one wavelength are used to compute the highest wavenumber component, the MDCD-HY will always be several times more accurate than WENO-JS according to Fig. 9.

4.2. The Shu–Osher problem

This problem is governed by the one-dimensional Euler equations with the following initial conditions:

$$(\rho,u,p)=\begin{cases}(3.857143,2.629369,10.3333), & \text{if } x\leqslant 1,\\ (1+0.2\sin(5x),0,1), & \text{otherwise.}\end{cases}$$

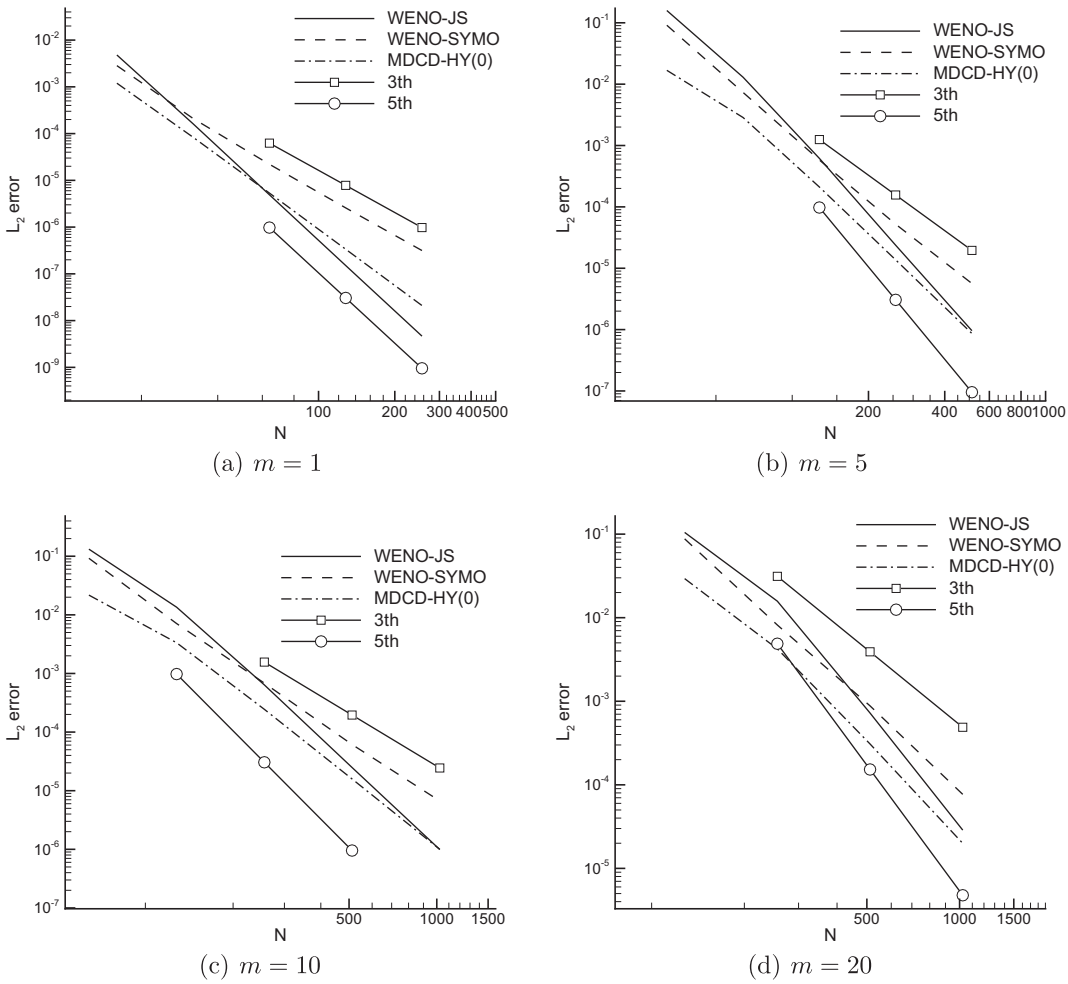


Fig. 9. The rate of convergence in terms of L_2 errors.

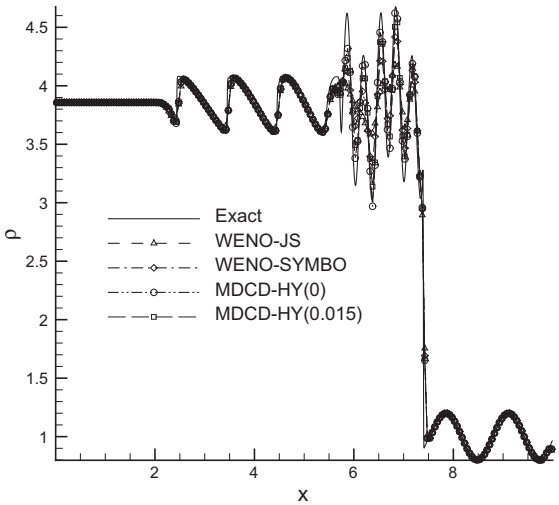


Fig. 10. Shu–Osher problems. Distribution of density at $t = 1.8$ for WENO-JS, WNO-SYMBO and MDCD-HY.

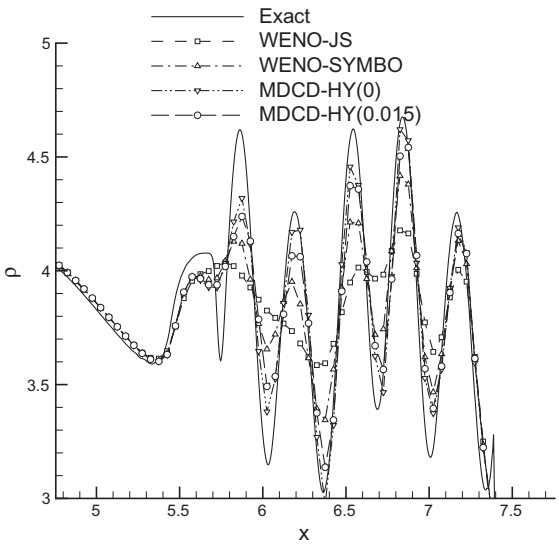


Fig. 11. The enlarged portion of Fig. 10 near the entropy disturbances.

This test case represents a Mach 3 shock wave interacting with a sine entropy wave which generates a flow field with both small scale structures and discontinuities. This is a good test case to evaluate the spectral properties as well as the shock-capturing capability of a scheme. The solution is advanced in time up to $t = 1.8$ on the computational domain $x \in [0, 10]$; the grid number is $N = 200$. Density distributions computed with different schemes are shown in Figs. 10 and 11. The solution obtained using WENO-JS on a $N = 2000$ grid is considered to be the "exact" solution although the real exact solution is not known for this problem. Both MDCD-HY(0) and MDCD-HY(0.015) demonstrate superior resolution than WENO-SYMBO and WENO-JS in reproducing the correct flow features downstream the shock wave. Because the dissipation of MDCD-HY(0.015) is similar to that of WENO-SYMBO, the improved dispersion property of MDCD-HY is responsible for its better resolution. Comparing results of MDCD-HY(0) with MDCD-HY(0.015), one can see that even a small amount of dissipation will influence the results. Therefore, the dissipation should be reduced as much as possible if it will not create spurious numerical oscillations. The flexibility in adjusting the numerical dissipation offered by MDCD-HY thus appears to be a great advantage for problems with both small scale structures and discontinuities.

4.3. Propagation of broadband sound waves

This test problem corresponds to the propagation of a sound wave packet which contains acoustic turbulent structures with various length scales. The initial conditions are given by

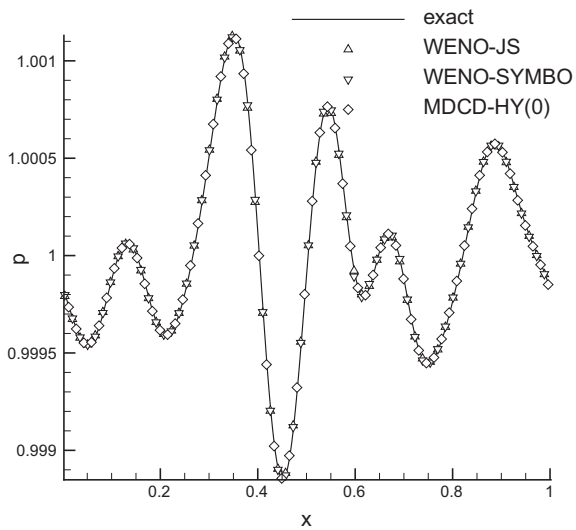


Fig. 12. Broadband wave propagation. Distribution of pressure for WENO-JS, WNO-SYMBO and MDCD-HY(0) with $k_0 = 4$.

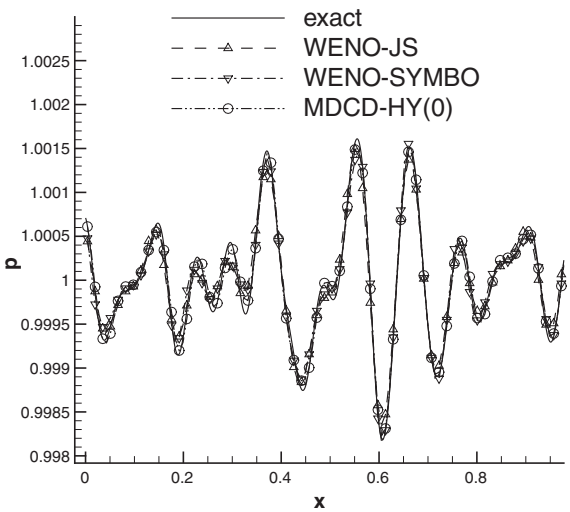


Fig. 13. Broadband wave propagation. Distribution of pressure for WENO-JS, WNO-SYMBO and MDCD-HY(0) with $k_0 = 8$.

$$p(x,0) = p_0\left(1 + \varepsilon \sum_{k=1}^{N/2} (E_p(k))^{0.5} \sin(2\pi k(x + \psi_k))\right),$$
$$\rho(x,0) = \rho_0 (p(x,0)/p_0)^{1/\gamma},$$
$$u(x,0) = u_0 + \tfrac{2}{\gamma-1}(c(x,0)/c_0),$$

where

$$E_p(k) = \left(\frac{k}{k_0}\right)^4 e^{-2(k/k_0)^2}$$

is the energy spectrum which reaches its maximum at $k = k_0$. The computation domain is $x \in [0,1]$ and ψ_k is a random number ranging from 0 to 1. ε is a small parameter which determines the intensity of the acoustic turbulence and is chosen to be $\varepsilon = 0.001$ in the simulation. $c = \sqrt{\frac{\gamma p}{\rho}}$ is the speed of sound. Periodic boundary conditions are imposed at the boundaries. Computations have been carried out with different schemes for $k_0 = 4$, 8, 12 on a $N = 128$ grid using a Courant number $CFL = 0.2$. The simulations are computed for one period of time. When $k_0 = 4$, most part of the turbulent energy is

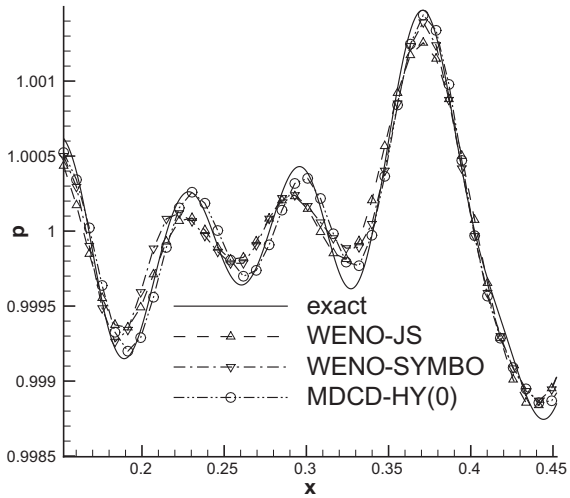


Fig. 14. A randomly chosen enlarge portion of Fig. 13.

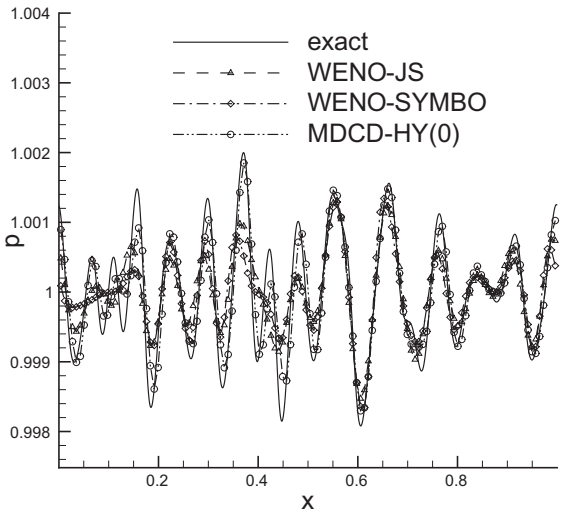


Fig. 15. Broadband wave propagation. Distribution of pressure for WENO-JS, WNO-SYMO and MDCD-HY(0) with $k_0 = 12$.

concentrated on low wavenumbers; when $k_0 = 12$, the high wavenumber structures are more energetic. The numerical results in terms of pressure distribution are shown in Figs. 12–16. When $k_0 = 4$, all schemes give very similar numerical results which means that all schemes are capable of capturing low wavenumber flow features. However, as k_0 increases, the differences in resolution among the various schemes become more obvious. MDCD-HY(0) performs remarkably better than WENO-JS and WENO-SYMBO according to Figs. 13–16.

4.4. Two-dimensional viscous shock tube

This test case is taken from [29]. A membrane, which separates two different states of the perfect gas, is initially located at the middle of a square tube with unit length. After the membrane broken, a shock wave, followed by a contact discontinuity, moves to the right. They interact with the horizontal wall and create a thin boundary layer. After the shock reflects back from the right wall, it interacts with the contact discontinuity and the boundary layer, producing complex flow structures. Several test cases with different initial condition are considered in the present paper to test the computational efficiency as well as the robustness of the proposed scheme. The non-dimensional initial conditions are shown in Table 3 where the cases A, B and C represent flows with increasingly strong shock waves. In the present paper, we choose $Re = 200$ and a 300×150 uniform grid for all test cases. The simulations are run to $t = 1.0$ with $CFL = 0.3$. For test case A, MDCD-HY(0), WENO-JS and WENO-SYMBO can give stable solutions and the numerical results depicted in Fig. 17 show similar resolution. For test case

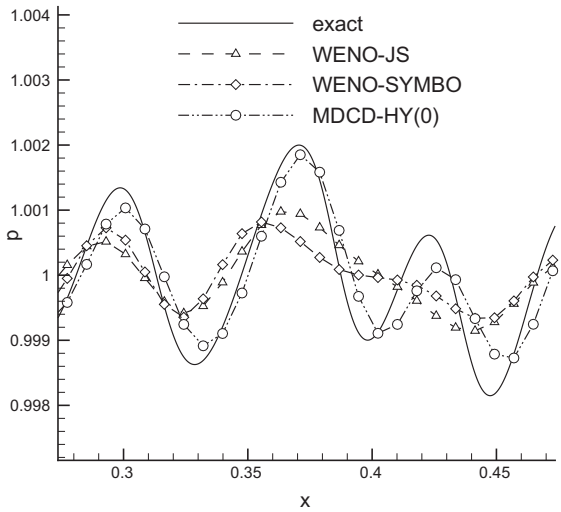


Fig. 16. A randomly chosen enlarge portion of Fig. 15.

Table 3
Physical parameters for the viscous shock tube problem.

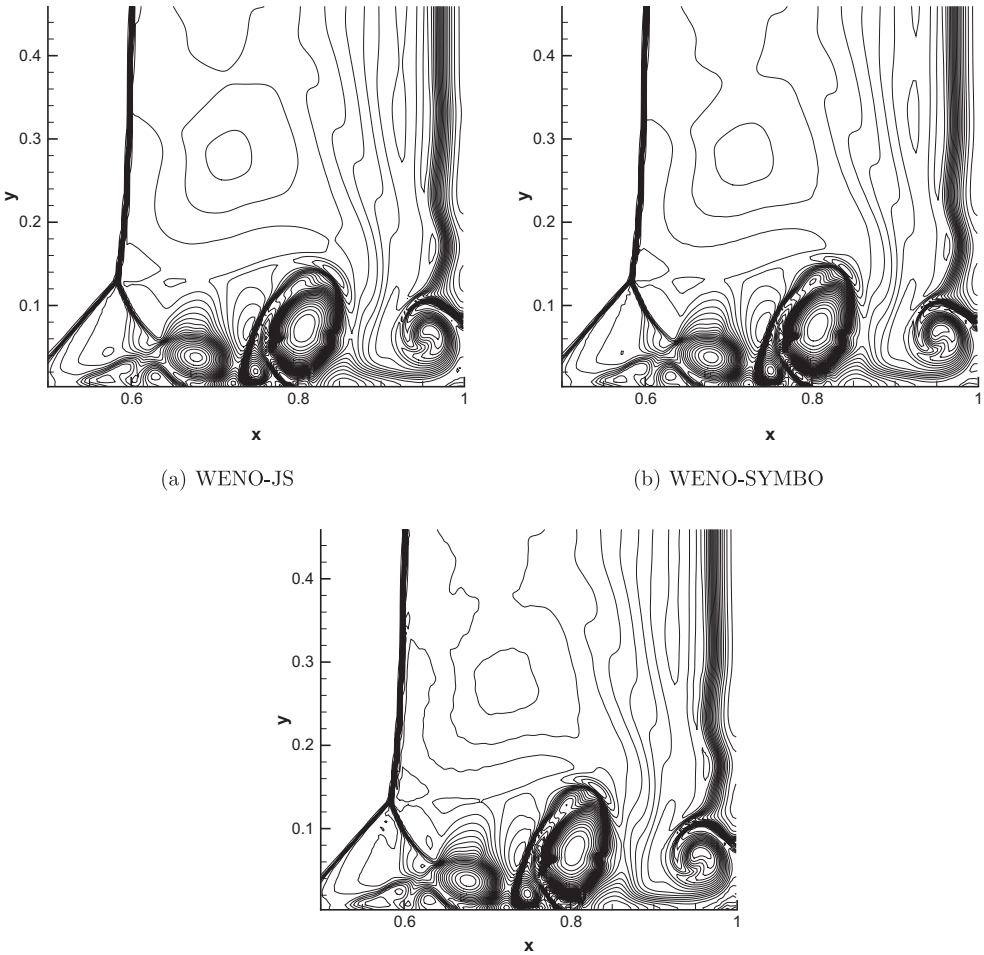
Case	ρ_L	u_L	v_L	P_L	ρ_R	u_R	v_R	P_R
A	120	0	0	ρ_L/γ	1.2	0	0	ρ_R/γ
B	240	0	0	ρ_L/γ	1.2	0	0	ρ_R/γ
C	480	0	0	ρ_L/γ	1.2	0	0	ρ_R/γ

B, the dissipation of WENO-SYMBO is too small and the computation breaks down. On the other hand, WENO-JS and MDCD-HY(0.02) are capable of supplying sufficient numerical dissipations. The resolution of MDCD-HY(0.02) is higher than WENO-JS in capturing the subtle flow structures as shown in Fig. 18. If the shock wave is made even stronger in terms of the initial conditions of case C, WENO-JS, WENO-SYMBO and MDCD-HY(0.02) become unstable while MDCD-HY(0.046) can still get stable numerical results because of its larger numerical dissipation. The corresponding numerical results are shown in Fig. 19. Therefore, the flexibility in adjusting the numerical dissipation makes MDCD-HY more robust. Considering computational efficiency, Table 4 clearly shows that MDCD-HY is the most efficient. Comparing with WENO-JS, MDCD-HY is more than 20% faster for case A and about 15% more efficient for case B.

4.5. Compressible wall turbulence

Compressible turbulent channel flow between isothermal walls is chosen as a practical test case to further investigate the performance of the MDCD-HY scheme. In the present paper, we consider a Mach 1.5 supersonic flow [1]. Some basic parameters of the simulation can be found in Table 5. x, y and z represent streamwise, spanwise and wall normal directions, respectively. The Reynolds number $Re_m = \rho_m U_m \delta/\mu_w$ is based on the bulk density, the bulk velocity, the channel half-width and viscosity at the isothermal wall; the Reynolds number $Re_\tau = u_\tau \delta/\nu$ (where u_τ is defined as $u_\tau = (\tau_w/\rho_w)^{1/2}$ and τ_w is the wall shear stress) is based on the friction velocity, the channel half-width, and the kinematic viscosity at the wall; the Mach number $Ma_m = U_m/a_w$ is based on the bulk velocity and the speed of sound at the isothermal wall. Pr is the Prandtl number and γ is the ratio of specific heat. Periodic boundary conditions are used in both streamwise direction and spanwise direction. As suggested by Coleman et al. [1], the flow is driven by a uniform body force rather than a pressure gradient to preserve streamwise homogeneity. The body force is chosen to vary in time so that the total mass flow rate remains constant.

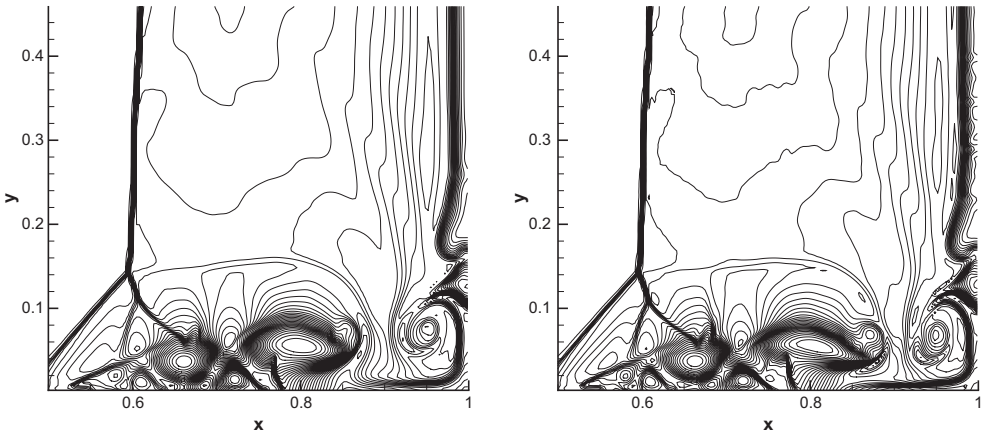
Although the flow is supersonic, no shock waves are found in the numerical solutions. For the flows without shock waves, Sandham et al. [30] found that if the spatial difference operators satisfied a summation-by-part condition, no numerical dissipation was needed. Based on this finding, they developed a stable scheme by an entropy-splitting approach. However, in our DNS using the proposed finite difference scheme, we find that MDCD-HY(0) is not able to supply enough numerical dissipation to stabilize the simulation. On the other hand, MDCD-HY(0.01) is stable and is thus implemented. We firstly apply MDCD-HY(0.01) on a 160^3 grid. A non-uniform mesh with a cosine distribution is employed in the wall-normal direction. The resulting mesh sizes are $\Delta x^+ = 14.1$, $\Delta y^+ = 7.0$ and $\Delta z^+ = 0.04 \sim 4.5$. The comparison of the computed statistics with the DNS data of Coleman et al. [1] is shown in Fig. 20. The agreements are satisfactory.



(a) WENO-JS (b) WENO-SYMBO

(c) MDCD-HY(0)

Fig. 17. Density contours for case A at $t = 1$ computed by WENO-JS, WENO-SYMBO and MDCD-HY(0).



(a) WENO-JS (b) WENO-HY(0.02)

Fig. 18. Density contours for case B at $t = 1$ computed by WENO-JS and MDCD-HY(0.02).

Secondly, the same flow fields are computed by using MDCD-HY(0.01) and WENO-JS on a 96^3 grid to study the dispersion and dissipation of the numerical schemes in terms of the under-resolved simulations. The streamwise one-dimensional

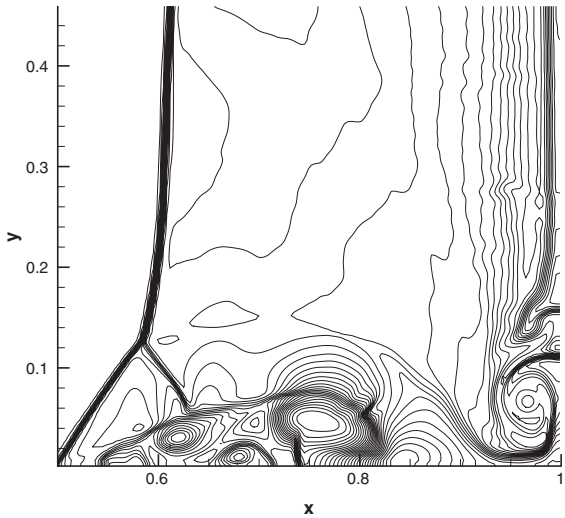


Fig. 19. Density contours for case C at $t = 1$ computed MDCD-HY(0.046).

Table 4
Computational time in seconds. "×" means that the scheme cannot get a converged result.

	Case A	Case B	Case C
WENO-JS	1080.7	1212.4	×
WENO-SYMBO	1134.9	×	×
MDCD-HY	861.8	1052.5	1157.6

Table 5
Parameters for compressible turbulent channel flow.

Re_m	Re_τ	Ma_m	Pr	γ	L_x	L_y	L_z	Δt
3000	220	1.5	0.72	1.4	$4\pi\delta$	$2\pi\delta$	2δ	$0.12\nu/u_\tau^2$

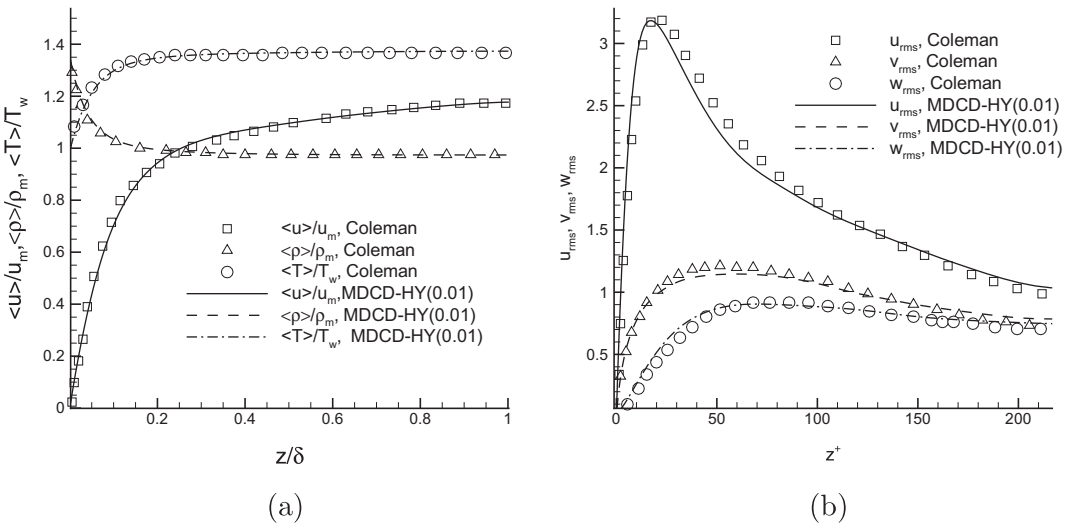
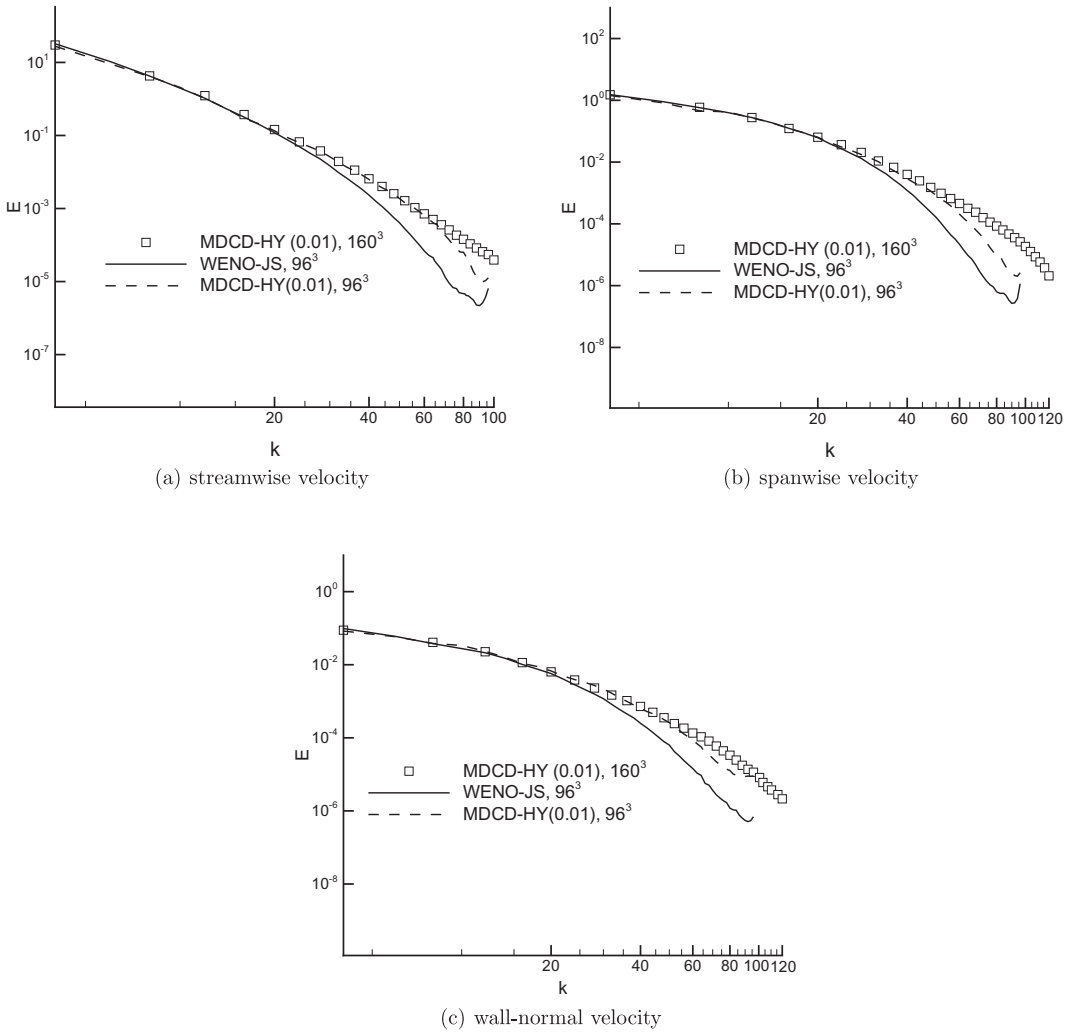


Fig. 20. The statistic quantities of DNS. (a) Mean streamwise velocity, mean density and mean temperature. (b) Root-mean-square velocity fluctuations normalized by friction velocity.

spectra at $z^+ = 10$ are shown in Fig. 21. For the low wavenumber components of the spectra, both MDCD-HY(0.01) and WENO-JS agree with the resolved simulation. However, when $k \geqslant 30$, the results of WENO-JS start to deviate from the fine



(a) streamwise velocity

(b) spanwise velocity

(c) wall-normal velocity

Fig. 21. One-dimensional velocity energy spectra at $z^+ = 10$.

grid results. For MDCD-HY(0.01), the deviation is not significant until $k \geqslant 60$. These results clearly show that MDCD-HY(0.01) is less dispersive and achieves higher resolution. It is also noted the larger numerical dissipation of WENO-JS produces larger drops of the spectra at higher wavenumbers.

5. Conclusions

In the present paper, a class of semi-discrete finite difference schemes with the MDCD properties has been designed. A remarkable feature of these schemes is the possibility to optimize the dissipation and dispersion properties independently. A sufficient condition for the finite difference schemes to have independent dispersion and dissipation is introduced and a 4th order MDCD scheme is constructed. In order to make the scheme capable of capturing shock waves, the MDCD strategy is then applied in the framework of WENO scheme to design the MDCD–WENO scheme. To eliminate unnecessary numerical dissipation in the smooth region and to improve the computational efficiency, a 4th order MDCD-HY scheme is proposed by blending the 4th order MDCD–WENO scheme with its linear counterpart. The spectral properties of the nonlinear hybrid scheme is studied numerically by computing the ADR using the approach developed by Pirozzoli. It is found that by choosing proper dispersion coefficients, the dispersion error of the hybrid scheme can be smaller than WENO-JS and WENO-SYMBO using the same stencils. At the same time, the dissipation of the hybrid scheme can be adjusted in a very large range. These properties make the MDCD-HY scheme very flexible, robust and less time-consuming. Several standard test cases have been implemented which show excellent resolution and robustness of the MDCD-HY scheme. For practical application, the MDCD-HY scheme has been applied to the DNS of compressible turbulent channel flow between isothermal walls. The results of the MDCD-HY scheme are in good agreement with those computed by the spectral method and are better than the results predicted by more dissipative WENO-JS scheme.

Acknowledgments

This work was supported by Projects 10932005 and 50910222 of NSFC.

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